COMP2711 / DISCRETE MATHEMATICAL TOOLS

Graphs

Past Paper Solutions

60	1	2016-25
verified exercises	topic groups	source papers

English solutions aligned with the supplied official answers.

Each answer has two parts: the original Official Answer and the existing Supplementary Explanation. Source PDF pages are retained for verification.

Generated 2 October 2026

All verified, in-scope questions linked to Graphs in the knowledge base.

Each question is included once, including cross-topic applications.

READING GUIDE

Contents

Graphs
60 exercises
G001-G060

How to use this booklet

Questions are grouped by their first Graphs subtopic, then ordered by difficulty and year (newest first). Each question appears once as G001-G060. Difficulty ranges from 1 (introductory) to 4 (challenging).

Official Answer reproduces the answer in the source paper, with its physical PDF page reference.

Supplementary Explanation retains the existing independently authored reasoning, including the stated result and derivation. The two parts remain separate throughout the booklet.

Source references use physical PDF pages. File links work with the local source collection at its existing location.

The answer provenance shown for each exercise retains its knowledge-base classification; attaching an official quotation does not reclassify a derived correction as official.

G001

Graphs

2025 Spring Final | Question 12(a)(ii) | Source page 16 | Difficulty 1
Q-2025S-FINAL-P12-a-ii | Answer provenance: official

> (a) [3 pts] Consider a simple undirected graph with 6 nodes. Briefly explain (in at most 3 lines) why it is impossible for the nodes to have the following degree sequences:
>
> (ii) $4, 4, 3, 2, 1, 1$

Official Answer

Source: physical PDF page 16 (COMP2711_S2025_Final.pdf).

(ii) The sum of the degrees is odd. So not possible.

Supplementary Explanation

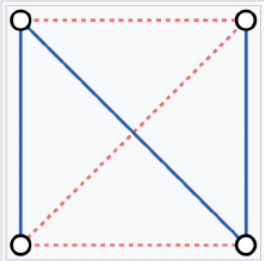
Answer. Impossible.

Derivation/proof. The proposed sum is $4 + 4 + 3 + 2 + 1 + 1 = 15$. The handshaking lemma gives $\sum_v \deg(v) = 2|E|$, an even integer, whereas fifteen is odd. This contradiction rules out the graph without any attempted drawing. The argument itself fits within the requested three-line limit.

G002

Graphs

2024 Fall Final | Question 11(a) | Source page 27 | Difficulty 1 | 1 mark
Q-2024F-FINAL-P11-a | Answer provenance: official

> The complement or inverse of a graph G is a graph H on the same vertices such that two distinct vertices of H are adjacent if and only if they are not adjacent in G. A self-complementary graph is a graph which is isomorphic to its complement. As an example, the following graph is self-complementary.
>
> 
>
>
> Assume a graph G is a self-complementary graph with n nodes. Consider H as its complement. Assume S to be the sum of all the nodes degrees in G.
>
> (a) Is S even or odd? No argument is needed.

Official Answer

Source: physical PDF page 27 (COMP2711_F2024_Final.pdf).

> Answer: (a) Even

Supplementary Explanation

Answer. Even.

Derivation/proof. Let $E(G)$ be the edge set. Each ordinary undirected edge contributes one to each of its two distinct endpoints, hence two to the degree sum. The complement definition uses adjacency between distinct vertices, so loops do not enter. The handshaking identity is $S = \sum_{v \in V(G)} \deg_G(v) = 2|E(G)|$. Since the number of edges is an integer, S is even, including the edgeless case $S = 0$.

G003

Graphs

2022 Final Part 2 | Question 5(a)(ii) | Source page 3 | Difficulty 1
Q-2022-FINAL2-P05-a-ii | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. You can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (i) Every edge connects a vertex in V_1 and a vertex in V_2 (ii) For every vertex in V_1, there are edges that connect it with all vertices in V_2
>
> For every vertex in V_1, there are edges that connect it with all vertices in V_2

Official Answer

Source: physical PDF page 3 (2022-final-part-2.pdf).

> (ii) $\forall u \in V_1 \forall v \in V_2 (E(u, v))$.

Supplementary Explanation

Answer. $\forall u \in V_1 \forall v \in V_2 E(u, v)$.

Derivation/proof. For each u in the first part, assert E(u,v) for every v in the second. This is exactly two restricted universal quantifiers. Undirectedness supplies the reverse orientation automatically. Since neither quantifier ranges over two vertices in the same part, the formula says nothing about internal edges.

Internal edges have not been forbidden.

G004

Graphs

2022 Final Part 2 | Question 5(b) | Source page 3 | Difficulty 1 | 2 marks
Q-2022-FINAL2-P05-b | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. You can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (i) Every edge connects a vertex in V_1 and a vertex in V_2 (ii) For every vertex in V_1, there are edges that connect it with all vertices in V_2
>
> If (a)(i) is true, is G necessarily a bipartite graph? Please give a brief justification.

Official Answer

Source: physical PDF page 3 (2022-final-part-2.pdf).

> (b) Yes. Based on the definition of bipartite graph, the graph G is bipartite because it can be partitioned into two disjoint subsets V_1 and V_2 and every edge connects a vertex in V_1 and a vertex in V_2.

Supplementary Explanation

Answer. Yes.

Derivation/proof. A bipartite graph is one whose vertices can be partitioned into two disjoint parts so every edge joins the parts. V1,V2 already give the disjoint covering partition, and (a)(i) gives exactly the edge restriction. Therefore they witness a bipartition of G. Completeness of cross edges is not part of this definition.

The assumptions $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$ make these sets a witness to a bipartition when every edge crosses.

G005

Graphs

2021 Final Part 2 | Question 5(d) | Source page 4 | Difficulty 1 | 2 marks
Q-2021-FINAL2-P05-d | Answer provenance: official

> For each of the following parts, either give an example graph or prove that such a graph does not exist.
>
> A directed graph with 5 vertices, v_1, v_2, v_3, v_4 and v_5.
>
> - In-degree of $v_1 = 1$, out-degree of $v_1 = 3$.
> - In-degree of $v_2 = 2$, out-degree of $v_2 = 1$.
> - In-degree of $v_3 = 3$, out-degree of $v_3 = 2$.
> - In-degree of $v_4 = 1$, out-degree of $v_4 = 1$.
> - In-degree of $v_5 = 2$, out-degree of $v_5 = 1$.

Official Answer

Source: physical PDF page 4 (2021-final-part-2.pdf).

> (d) None. The sum of in-degrees is different with the sum of out-degrees.

Supplementary Explanation

Answer. No such directed graph exists.

Derivation/proof. Each directed edge contributes exactly one to total outdegree and exactly one to total indegree, including a loop if loops were allowed. Here indegrees sum to 1+2+3+1+2=9, but outdegrees sum to 3+1+2+1+1=8. They would both have to equal the number of edges, so the incompatible totals rule out every graph convention.

The contradiction is that both totals must equal the number of edges.

G006

Graphs

2020 Final Part 1 | Question 1(a)(iii) | Source page 1 | Difficulty 1
Q-2020-FINAL1-P01-a-iii | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \Leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. Your can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (ii) "Every edge connects a vertex in V_1 and a vertex in V_2".
>
> (iii) "For every vertex in V_1, there are edges that connect it with all vertices in V_2".
>
> "For every vertex in V_1, there are edges that connect it with all vertices in V_2".

Official Answer

Source: physical PDF page 1 (2020-final-part-1.pdf).

$\forall u \in V_1 \forall v \in V_2 (E(u, v))$.

Supplementary Explanation

Answer. $\forall u \in V_1 \forall v \in V_2 E(u, v)$.

Derivation/proof. The phrase "every vertex" in the first part supplies the first universal quantifier; "all vertices" in the other part supplies the second. For each resulting pair assert adjacency. This places no restriction on edges whose two endpoints are both in V1 or both in V2.

The formula $\forall u \in V_1 \forall v \in V_2 E(u, v)$ requires all cross-part edges. By itself it does not prohibit edges within a part; that is a separate restriction in the preceding property.

G007

Graphs

2020 Final Part 1 | Question 1(b) | Source page 1 | Difficulty 1 | 2 marks
Q-2020-FINAL1-P01-b | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \Leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. Your can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (ii) "Every edge connects a vertex in V_1 and a vertex in V_2".
>
> (iii) "For every vertex in V_1, there are edges that connect it with all vertices in V_2".
>
> If (a)(ii) is true, is G necessarily a bipartite graph? Your answer should be either "Yes" or "No". Justification is not necessary.

Official Answer

Source: physical PDF page 1 (2020-final-part-1.pdf).

Yes.

Supplementary Explanation

Answer. Yes.

Derivation/proof. The source gives disjoint parts V1,V2 whose union is V. Condition (a)(ii) ensures every edge has one endpoint in each part. That is precisely a bipartition of G, so G is bipartite. Missing some cross edges would affect completeness but not bipartiteness.

The sets must satisfy $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$ as well as contain no internal edges. These conditions make the displayed partition a valid bipartition, rather than merely two independent subsets omitting some vertices.

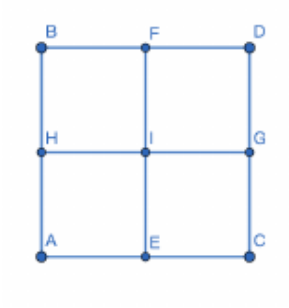
G008

Graphs

2020 Final Part 2 | Question 4(c) | Source page 3 | Difficulty 1
Q-2020-FINAL2-P04-c-decision | Answer provenance: official

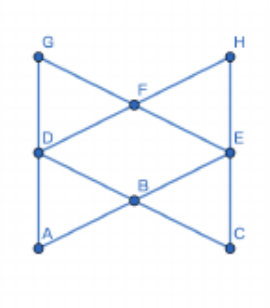
For each of the following graphs, determine whether it has an Euler cycle. If not, insert the least number of edges so that it has an Euler cycle. (You can either draw a picture or write down which edges should be inserted.)

(a) Figure 1



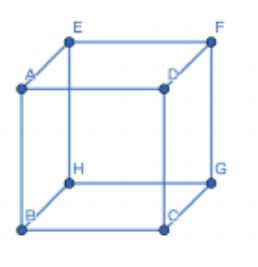
(a) Figure 1

(b) Figure 2



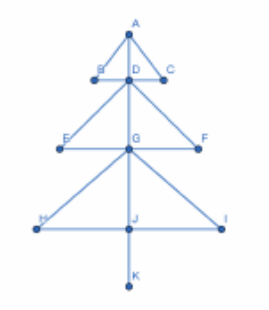
(b) Figure 2

(c) Figure 3



(c) Figure 3

(d) Figure 4



(d) Figure 4

(c) Figure 3 — determine whether it has an Euler cycle.

Official Answer

Source: physical PDF page 3 (2020-final-part-2.pdf).

It is not an Euler circuit, since all vertices have odd degree.

Supplementary Explanation

Answer. No.

Derivation/proof. Figure 3 is the cube graph on A,B,C,D,E,F,G,H. At each vertex exactly three cube edges meet, so all eight vertices have odd degree 3. Although the graph is connected, the Euler cycle condition requires all degrees even and therefore fails at every vertex.

In the cube drawing all eight degrees are 3. The unmarked crossings do not add vertices, so they cannot change the count or repair its eight odd degrees.

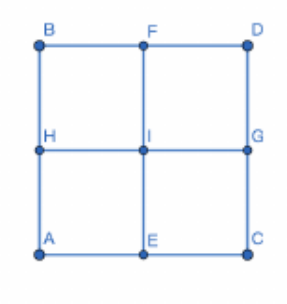
G009

Graphs

2020 Final Part 2 | Question 4(d) | Source page 3 | Difficulty 1
Q-2020-FINAL2-P04-d-insert | Answer provenance: official

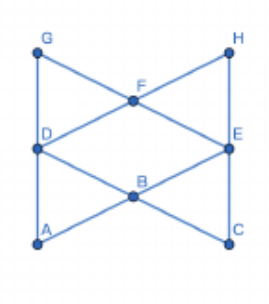
For each of the following graphs, determine whether it has an Euler cycle. If not, insert the least number of edges so that it has an Euler cycle. (You can either draw a picture or write down which edges should be inserted.)

(a) Figure 1



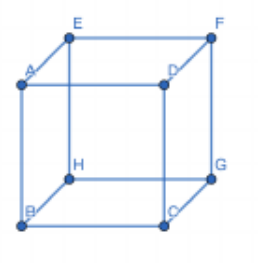
(a) Figure 1

(b) Figure 2



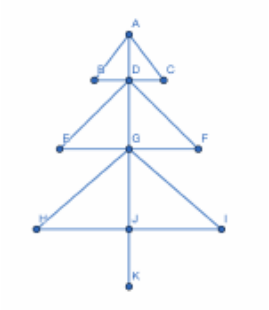
(b) Figure 2

(c) Figure 3



(c) Figure 3

(d) Figure 4



(d) Figure 4

(d) Figure 4 — if not, insert the least number of edges so that it has an Euler cycle.

Official Answer

Source: physical PDF page 3 (2020-final-part-2.pdf).

It is not an Euler circuit, since A, K have odd degree. The official answer drawing adds a curved edge AK outside the graph.

Supplementary Explanation

Answer. Insert AK; the minimum is 1 edge.

Derivation/proof. Only A and K have odd degree, respectively 3 and 1. At least one edge must be added because the original graph has no Euler cycle. The absent edge AK increases those degrees to 4 and 2, changing no other parity. The graph remains connected, so all Euler conditions hold after exactly one insertion.

The curved edge avoids creating incidental vertices. It changes A from degree three to four and K from degree one to two.

Adding AK changes the degree of A from three to four and that of K from one to two. All other degrees stay even, so one edge is minimal and sufficient.

G010

Graphs

2018 Final | Question 4(c) | Source page 6 | Difficulty 1
Q-2018-FINAL-P04-c-basis | Answer provenance: official

> For this question, no proof is required. Fill in the following recursive definition so that S is the set of complete graphs.
>
> Recursive step: Suppose that $G = (V, E)$ is in S, where $V = \{v_1, v_2, \ldots, v_n\}$. Then $G' = (V', E')$ is in S, where $V' = \{v_1, v_2, \ldots, v_n, v_{n+1}\}$.
>
> (c) Fill in the following recursive definition so that S is the set of complete graphs.
>
> Basis: ____ is in S.
>
> [Atomic request: basis graph; no proof is required.]

Official Answer

Source: physical PDF page 6 (2018-final.pdf).

Basis: The graph with one vertex and no edges is a complete graph.

Supplementary Explanation

Answer. K_1, the graph with one vertex and no edges.

Derivation/proof. A single vertex has no pair of distinct vertices, so the condition that every such pair is adjacent holds vacuously. It is therefore a complete graph. Adding one vertex at a time from this basis can reach every positive number of vertices, whereas starting with K_2 would omit K_1.

Start with a single vertex with no edges. Adding one new vertex adjacent to every old vertex successively constructs $K_2, K_3, \ldots$; no self-loop is added.

G011

Graphs

2017 Final | Question 14(a) | Source page 18 | Difficulty 1 | 1 mark
Q-2017-FINAL-P14-a | Answer provenance: official

> In this question, we want to find the minimum number of edges in a connected graph with n vertices.
>
> (a) Give an example of a connected graph with n vertices and $n - 1$ edges.

Official Answer

Source: physical PDF page 18 (2017-final.pdf).

Remove any one edge from C_n.

Supplementary Explanation

Answer. The path $v_1 - v_2 - \cdots - v_n$.

Derivation/proof. Use vertices $v_1, \ldots, v_n$ and edges $\{v_i, v_{i+1}\}$ for $1 \leq i < n$. There are exactly $n - 1$ edges. Any pair v_i, v_j is joined by the consecutive segment of this path, so the graph is connected. For $n = 1$, the singleton with no edges has the required properties. For $n \geq 3$, this is the source's cycle with one edge removed.

G012

Graphs

2025 Fall Final | Question 11(a)(i) | Source page 19 | Difficulty 2 | 1 mark
Q-2025F-FINAL-P11-a-i | Answer provenance: official

> (a) [3 points, 1 each] Determine whether the following statements are True or False. No argument is needed.
>
> (i) A simple graph is bipartite if and only if it contains at least one odd cycle.

Official Answer

Source: physical PDF page 19 (COMP2711_F2025_Final.pdf).

(a) (i) False

Supplementary Explanation

Answer. False.

Derivation/proof. In a bipartition $V = L \cup R$, every edge joins opposite sides. Along a cycle, the side alternates at each step. After an odd number of steps one is on the opposite side from the start, so an odd cycle cannot return to its starting vertex. A bipartite graph therefore has no odd cycle, contrary to the printed statement's condition. A single-edge graph is a concrete counterexample: it is bipartite and has no cycle. Conversely, a triangle has an odd cycle and is not bipartite.

G013

Graphs

2025 Spring Final | Question 12(a)(i) | Source page 16 | Difficulty 2
Q-2025S-FINAL-P12-a-i | Answer provenance: official

> (a) [3 pts] Consider a simple undirected graph with 6 nodes. Briefly explain (in at most 3 lines) why it is impossible for the nodes to have the following degree sequences:
>
> (i) $5, 4, 3, 2, 2, 0$

Official Answer

Source: physical PDF page 16 (COMP2711_S2025_Final.pdf).

(a) (i) One node has a degree of 5 (means its connected to every other node) and one node has a degree of 0 (meanings not connected to anything) which is a contradiction.

Supplementary Explanation

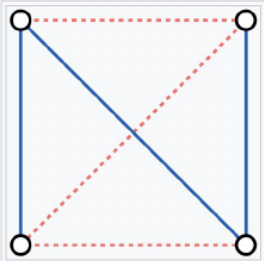
Answer. Impossible.

Derivation/proof. Suppose such a simple undirected graph exists and take a degree-five vertex v. There are six vertices and no loops or parallel edges, so its five neighbors are all the other vertices. In particular, it is adjacent to the asserted degree-zero vertex w, forcing $\deg(w) \geq 1$, a contradiction. In the requested three-line form: a degree-five vertex must connect to the degree-zero vertex, which would cease to be isolated.

G014

Graphs

2024 Fall Final | Question 11(b) | Source page 27 | Difficulty 2 | 3 marks
Q-2024F-FINAL-P11-b | Answer provenance: official

> The complement or inverse of a graph G is a graph H on the same vertices such that two distinct vertices of H are adjacent if and only if they are not adjacent in G. A self-complementary graph is a graph which is isomorphic to its complement. As an example, the following graph is self-complementary.
>
> 
>
> Graph A is isomorphic to its complement.
>
> Assume a graph G is a self-complementary graph with n nodes. Consider H as its complement. Assume S to be the sum of all the nodes degrees in G.
>
> (b) Argue why $2S = n(n-1)$? Hint: Two isomorphic graphs have same number of edges.

Official Answer

Source: physical PDF page 27 (COMP2711_F2024_Final.pdf).

> (b) Note that G and H are isomorphism, the sum of the degree of the nodes in these two graphs added is equal to $2S$. At the same time, note that for every node v, $d_v^G + d_v^H = n - 1$. So the sum of the degree of all nodes together in both graph is equal to $n(n-1)$.

Supplementary Explanation

Answer. $2S = n(n-1)$.

Derivation/proof. Self-complementarity gives $G \cong H$, so the graphs have equal edge counts. The handshaking identity yields $\sum_v \deg_H(v) = 2|E(H)| = 2|E(G)| = S$. For fixed v, each of the other $n-1$ vertices is adjacent to it in exactly one graph, giving $\deg_G(v) + \deg_H(v) = n - 1$. Summing over all n vertices gives $S + S = 2S$ on the left and $n(n-1)$ on the right.

G015

Graphs

2022 Final Part 2 | Question 5(a)(i) | Source page 3 | Difficulty 2
Q-2022-FINAL2-P05-a-i | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. You can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (i) Every edge connects a vertex in V_1 and a vertex in V_2 (ii) For every vertex in V_1, there are edges that connect it with all vertices in V_2
>
> Every edge connects a vertex in V_1 and a vertex in V_2

Official Answer

Source: physical PDF page 3 (2022-final-part-2.pdf).

> (a) (i) $(\forall u \in V_1 \forall v \in V_1 (\neg E(u, v))) \wedge (\forall u \in V_2 \forall v \in V_2 (\neg E(u, v)))$ or $\forall u, v \in V (E(u, v) \rightarrow ((u \in V_1 \wedge v \in V_2) \vee (u \in V_2 \wedge v \in V_1)))$

Supplementary Explanation

Answer. $(\forall u \in V_1 \forall v \in V_1 \neg E(u, v)) \wedge (\forall u \in V_2 \forall v \in V_2 \neg E(u, v))$.

Derivation/proof. Because V1,V2 partition all vertices, any pair either lies within one part or across the two. The formula excludes both within-part possibilities, leaving only cross-part pairs as potential edges. Conversely every cross-part-only graph satisfies both universal nonadjacency clauses. It places no requirement that any particular cross edge exist.

The formula does not require every cross-part pair to be adjacent.

G016

Graphs

2022 Final Part 2 | Question 5(c) | Source page 3 | Difficulty 2 | 2 marks
Q-2022-FINAL2-P05-c | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. You can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (i) Every edge connects a vertex in V_1 and a vertex in V_2 (ii) For every vertex in V_1, there are edges that connect it with all vertices in V_2
>
> If (a)(ii) is true, is G necessarily a complete bipartite graph? Please give a brief justification.

Official Answer

Source: physical PDF page 3 (2022-final-part-2.pdf).

> (c) No. If there are edges connecting two vertices in V_1, the graph G is not a bipartite graph, so it is not a complete bipartite graph.

Supplementary Explanation

Answer. No.

Derivation/proof. Use V1={a,b}, V2={c}, and the triangle edges ac,bc,ab. Every vertex of V1 is connected to all vertices of V2, so (a)(ii) holds. The extra internal edge ab is not forbidden. It creates an odd cycle, so this graph is not bipartite at all, and hence not complete bipartite. This is a simple undirected counterexample.

Take $V_1 = \{a, b\}$ and $V_2 = \{c\}$ with edges ac, bc, ab: the triangle K_3. The edge ab forces a, b into opposite parts of any proposed bipartition, but c must be opposite both, impossible. To see why failure of the original partition alone is insufficient, take $V_1 = \{a, b\}$ and $V_2 = \varnothing$ with the sole edge ab. The cross-edge condition is vacuous, yet another partition $\{a\}, \{b\}$ makes this graph bipartite.

G017

Graphs

2022 Final Part 2 | Question 6(a) | Source page 3 | Difficulty 2
Q-2022-FINAL2-P06-a-expectation | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the expectation of X?

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (a) X_e is the indicator random variable that indicates the event of a pair of vertices $e = (u, v)$ having an edge. X_e is a Bernoulli trial, thus $E(X_e) = p$, $V(X_e) = p(1-p)$.
>
> There are $\binom{n}{2}$ pairs of vertices in total. Thus $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $E[X] = \binom{n}{2}p$.

Derivation/proof. Index the unordered pairs by e and define I_e=1 for an existing edge and 0 otherwise. The edge count is X=∑I_e. Each I_e has mean p. Summing by linearity over N=n(n−1)/2 pairs gives E[X]=Np. This matches the binomial mean for X distributed as Binomial(N,p); linearity alone already suffices for this calculation.

Write $X = \sum_e I_e$. Since $E[I_e] = 1 \cdot p + 0 \cdot (1-p) = p$,

$$E[X] = \sum_e p = Np = \binom{n}{2}p.$$

Linearity itself does not require independence.

G018

Graphs

2022 Final Part 2 | Question 6(a) | Source page 3 | Difficulty 2
Q-2022-FINAL2-P06-a-variance | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the variance of X?

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (a) X_e is the indicator random variable that indicates the event of a pair of vertices $e = (u, v)$ having an edge. X_e is a Bernoulli trial, thus $E(X_e) = p$, $V(X_e) = p(1-p)$.
>
> There are $\binom{n}{2}$ pairs of vertices in total. Thus $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $\mathrm{Var}(X) = \binom{n}{2}p(1-p)$.

Derivation/proof. For one edge indicator I, E[I²]=E[I]=p, so Var(I)=p−p². In the stated random graph distinct edge indicators are independent, giving covariance zero for every distinct pair. The variance of their sum is therefore N(p−p²), with N=C(n,2). At p=0 or1 the answer is zero, correctly reflecting a deterministic graph.

Because $I_e^2 = I_e$, its variance is $p - p^2 = p(1-p)$. Independence makes the covariance zero for each distinct pair; hence

$$\mathrm{Var}(X) = \sum_e \mathrm{Var}(I_e) = \binom{n}{2}p(1-p).$$

Equal marginal probabilities alone would not justify dropping covariance terms.

G019

Graphs

2021 Final Part 2 | Question 5(a) | Source page 3 | Difficulty 2 | 2 marks
Q-2021-FINAL2-P05-a | Answer provenance: official

> For each of the following parts, either give an example graph or prove that such a graph does not exist.
>
> A simple undirected graph with 10 vertices, whose degrees are $2, 1, 0, 3, 8, 9, 7, 2, 1, 1$.

Official Answer

Source: physical PDF page 4 (2021-final-part-2.pdf).

> (a) None. It is not possible to have both a vertex of degree 9 and a vertex of degree 0.

Supplementary Explanation

Answer. No such graph exists.

Derivation/proof. A degree-9 vertex in a simple graph on ten vertices is adjacent to all nine other vertices. In particular it must be adjacent to the listed degree-0 vertex. But a degree-0 vertex has no incident edges, contradicting that adjacency. This local contradiction suffices even without testing the entire degree sequence.

The degree-nine vertex reaches every other vertex; in particular, the listed degree-zero vertex must be adjacent to it. That vertex having no incident edge contradicts the forced edge.

G020

Graphs

2021 Final Part 2 | Question 5(b) | Source page 3 | Difficulty 2 | 2 marks
Q-2021-FINAL2-P05-b | Answer provenance: official

> For each of the following parts, either give an example graph or prove that such a graph does not exist.
>
> A complete bipartite undirected graph with 10 vertices, whose degrees are 3, 3, 3, 7, 7, 7, 7, 7, 7, 7.

Official Answer

Source: physical PDF page 4 (2021-final-part-2.pdf).

> (b) None. In a complete bipartite graph, we can partition the vertices into two sets V_1, V_2 so that there is an edge from every vertex in V_1 to every vertex in V_2. So, the degree of every vertex in V_1 should be $|V_2|$ and the degree of every vertex in V_2 should be $|V_1|$. When the two possible degrees are 3 and 7, we need 7 vertices with degree 3 and 3 vertices with degree 7 to make a complete bipartite graph. The given degrees are not in this case.

Supplementary Explanation

Answer. No such graph exists.

Derivation/proof. In a complete bipartite graph K_(a,b), each of the a vertices has degree b and each of the b vertices degree a. The only displayed degrees are 3 and 7, so the part sizes would have to be 3 and 7. Then there must be three degree-7 vertices and seven degree-3 vertices. The supplied sequence has the opposite multiplicities, so it cannot be the required graph.

The required sequence has three degree-seven vertices and seven degree-three vertices, but the printed sequence has three degree-three vertices and seven degree-seven vertices: the multiplicities are reversed.

G021

Graphs

2021 Final Part 2 | Question 5(c) | Source page 3 | Difficulty 2 | 2 marks
Q-2021-FINAL2-P05-c | Answer provenance: official

> For each of the following parts, either give an example graph or prove that such a graph does not exist.
>
> A strongly connected directed graph with 5 vertices, v_1, v_2, v_3, v_4 and v_5.
>
> - In-degree of $v_1 = 1$, out-degree of $v_1 = 3$.
> - In-degree of $v_2 = 2$, out-degree of $v_2 = 1$.
> - In-degree of $v_3 = 3$, out-degree of $v_3 = 1$.
> - In-degree of $v_4 = 0$, out-degree of $v_4 = 2$.
> - In-degree of $v_5 = 2$, out-degree of $v_5 = 1$.

Official Answer

Source: physical PDF page 4 (2021-final-part-2.pdf).

> (c) None. There is no path from v_i to v_4 for $i \neq 4$, because in-degree of v_4 is 0.

Supplementary Explanation

Answer. No such strongly connected graph exists.

Derivation/proof. Strong connectivity requires a directed path from every other vertex to v4. Such a path must end with an edge entering v4. Its indegree is prescribed as zero, so no entering edge exists and no such path can exist. The nonzero outdegree only permits leaving v4 and does not resolve the missing reverse reachability.

For $i \neq 4$, the last edge must enter v_4, but indegree zero forbids such an edge.

G022

Graphs

2020 Final Part 1 | Question 1(a)(ii) | Source page 1 | Difficulty 2
Q-2020-FINAL1-P01-a-ii | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \Leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. Your can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (ii) "Every edge connects a vertex in V_1 and a vertex in V_2".
>
> (iii) "For every vertex in V_1, there are edges that connect it with all vertices in V_2".
>
> "Every edge connects a vertex in V_1 and a vertex in V_2".

Official Answer

Source: physical PDF page 1 (2020-final-part-1.pdf).

$(\forall u \in V_1 \forall v \in V_1 (\neg E(u, v))) \wedge (\forall u \in V_2 \forall v \in V_2 (\neg E(u, v)))$.

Supplementary Explanation

Answer. $\forall u, v \in V (E(u, v) \rightarrow ((u \in V_1 \wedge v \in V_2) \vee (u \in V_2 \wedge v \in V_1)))$.

Derivation/proof. Universally consider both endpoints of an edge. Because the graph is undirected either orientation of the two parts is possible, so the consequent is a disjunction of the two opposite-part assignments. The given partition covers V and is disjoint, making this equivalent to forbidding edges within each part.

The two within-part restrictions are $\forall u \in V_1 \forall v \in V_1 \neg E(u, v)$ and $\forall u \in V_2 \forall v \in V_2 \neg E(u, v)$. These forbid every internal edge because the edge is undirected; this holds regardless of which endpoint is written first.

G023

Graphs

2020 Final Part 1 | Question 1(c) | Source page 1 | Difficulty 2 | 2 marks
Q-2020-FINAL1-P01-c | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \Leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. Your can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (ii) "Every edge connects a vertex in V_1 and a vertex in V_2".
>
> (iii) "For every vertex in V_1, there are edges that connect it with all vertices in V_2".
>
> If (a)(iii) is true, is G necessarily a complete bipartite graph? Your answer should be either "Yes" or "No". Justification is not necessary.

Official Answer

Source: physical PDF page 1 (2020-final-part-1.pdf).

No.

Supplementary Explanation

Answer. No.

Derivation/proof. Take $V_1 = \{a, b\}$, $V_2 = \{c\}$ and edges ac,bc,ab. Both required cross edges exist, so (a)(iii) holds, but the internal edge ab forms a triangle. A triangle cannot be bipartitioned and hence G is not a complete bipartite graph. This counterexample obeys simplicity and undirectedness.

The counterexample respects all stated premises: it is simple and undirected, and its two parts are disjoint and cover the vertices.

The internal edge *ab* violates the no-within-part-edge condition, even though every required cross-part edge is present. Thus the cross-edge property does not imply bipartiteness with that partition.

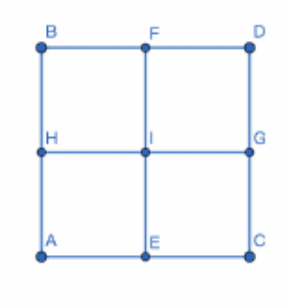
G024

Graphs

2020 Final Part 2 | Question 4(a) | Source page 3 | Difficulty 2
Q-2020-FINAL2-P04-a-decision | Answer provenance: official

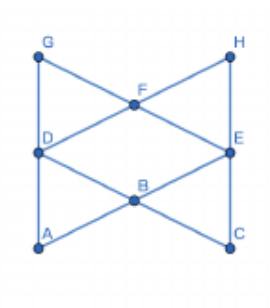
For each of the following graphs, determine whether it has an Euler cycle. If not, insert the least number of edges so that it has an Euler cycle. (You can either draw a picture or write down which edges should be inserted.)

(a) Figure 1



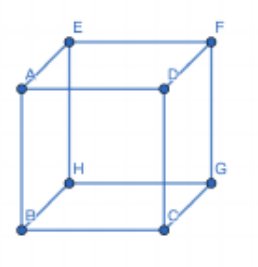
(a) Figure 1

(b) Figure 2



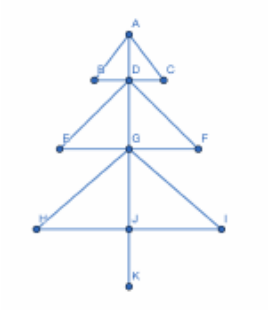
(b) Figure 2

(c) Figure 3



(c) Figure 3

(d) Figure 4



(d) Figure 4

(a) Figure 1 — determine whether it has an Euler cycle.

Official Answer

Source: physical PDF page 3 (2020-final-part-2.pdf).

It is not an Euler circuit, since E, H, G, F have odd degree.

Supplementary Explanation

Answer. No.

Derivation/proof. The graph is connected. Its four corners A,B,C,D have degree 2, its center I has degree 4, and E,F,G,H each have degree 3. A connected undirected graph has an Euler cycle exactly when every vertex has even degree. The four odd degrees therefore rule out an Euler cycle.

There are four odd-degree vertices, namely the four degree-three midpoints. The Euler-cycle criterion requires no odd-degree vertices, so the graph fails it despite being connected.

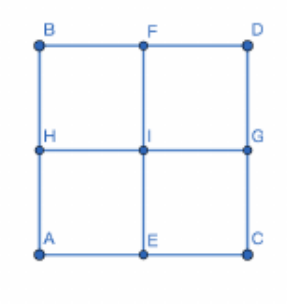
G025

Graphs

2020 Final Part 2 | Question 4(a) | Source page 3 | Difficulty 2
Q-2020-FINAL2-P04-a-insert | Answer provenance: official

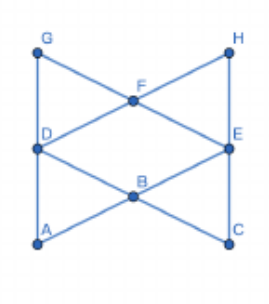
For each of the following graphs, determine whether it has an Euler cycle. If not, insert the least number of edges so that it has an Euler cycle. (You can either draw a picture or write down which edges should be inserted.)

(a) Figure 1



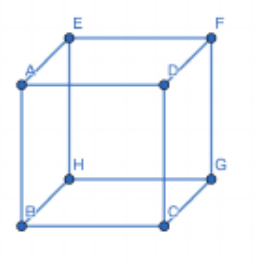
(a) Figure 1

(b) Figure 2



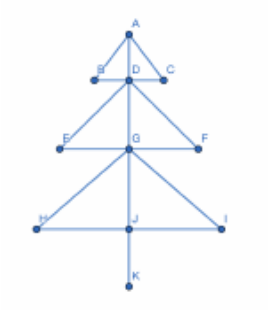
(b) Figure 2

(c) Figure 3



(c) Figure 3

(d) Figure 4



(d) Figure 4

(a) Figure 1 — if not, insert the least number of edges so that it has an Euler cycle.

Official Answer

Source: physical PDF page 3 (2020-final-part-2.pdf).

It is not an Euler circuit, since E, H, G, F have odd degree. The official answer drawing adds HF and EG.

Supplementary Explanation

Answer. Insert HF and EG; the minimum is 2 edges.

Derivation/proof. The odd vertices are E,F,G,H. Each inserted edge changes parity at at most two vertices, so at least four divided by two, or two edges, are necessary. HF and EG are absent in the original grid and pair all four odd vertices. After adding them all degrees are even and the graph remains connected, so the Euler criterion proves sufficiency and minimality.

The drawing is transcribed into edge names above; the original figure is on the cited physical page. Each odd midpoint changes from degree three to degree four.

An inserted edge can correct at most two odd degrees, so $4/2 = 2$ edges are necessary. The chosen pairs each raise their endpoint degrees from three to four and together attain this lower bound.

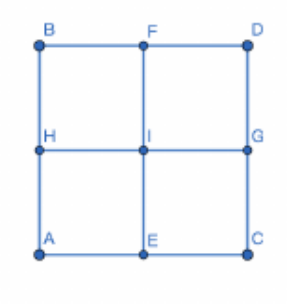
G026

Graphs

2020 Final Part 2 | Question 4(b) | Source page 3 | Difficulty 2 | 2 marks
Q-2020-FINAL2-P04-b | Answer provenance: official

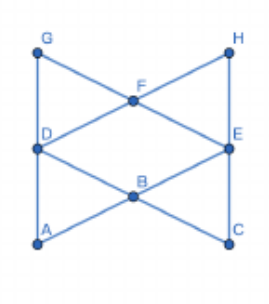
For each of the following graphs, determine whether it has an Euler cycle. If not, insert the least number of edges so that it has an Euler cycle. (You can either draw a picture or write down which edges should be inserted.)

(a) Figure 1



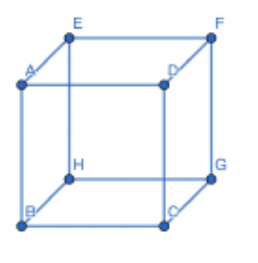
(a) Figure 1

(b) Figure 2



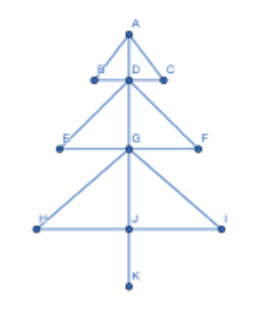
(b) Figure 2

(c) Figure 3



(c) Figure 3

(d) Figure 4



(d) Figure 4

(b) Figure 2 — determine whether it has an Euler cycle.

Official Answer

Source: physical PDF page 3 (2020-final-part-2.pdf).

It is an Euler circuit, since all vertices have even degree. The official answer drawing leaves Figure 2 unchanged.

Supplementary Explanation

Answer. Yes; no inserted edge is needed.

Derivation/proof. Read the labeled crossing vertices as part of the graph: B,F,D,E each have degree 4 and A,C,G,H each have degree 2. The graph is connected through D,E and the crossing vertices. Thus every vertex has even degree and the connected Euler criterion guarantees an Euler cycle.

Since the graph is already connected with all degrees even, zero insertions are necessary. This is both feasible and minimal.

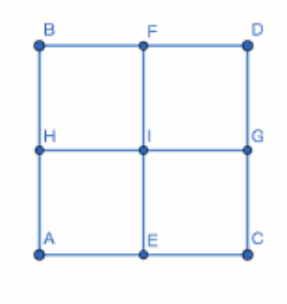
G027

Graphs

2020 Final Part 2 | Question 4(c) | Source page 3 | Difficulty 2
Q-2020-FINAL2-P04-c-insert | Answer provenance: official

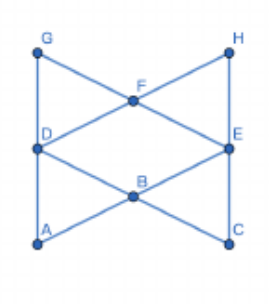
For each of the following graphs, determine whether it has an Euler cycle. If not, insert the least number of edges so that it has an Euler cycle. (You can either draw a picture or write down which edges should be inserted.)

(a) Figure 1



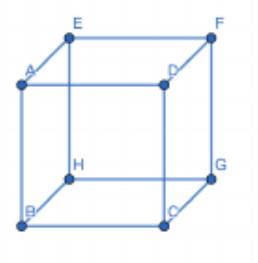
(a) Figure 1

(b) Figure 2



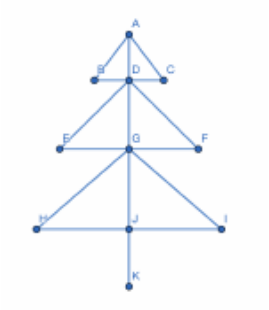
(b) Figure 2

(c) Figure 3



(c) Figure 3

(d) Figure 4



(d) Figure 4

(c) Figure 3 — if not, insert the least number of edges so that it has an Euler cycle.

Official Answer

Source: physical PDF page 3 (2020-final-part-2.pdf).

It is not an Euler circuit, since all vertices have odd degree. The official answer drawing adds AH, BE, DG, CF.

Supplementary Explanation

Answer. Insert AH, BE, DG, CF; the minimum is 4 edges.

Derivation/proof. All eight cube vertices have degree 3, so at least four inserted edges are needed to change eight odd parities. The four listed pairs are nonedges of the original cube and use every vertex once. They raise every degree to 4; connectivity is preserved. Thus the resulting graph has an Euler cycle, attaining the parity lower bound.

The drawing is transcribed into edge names above; every vertex occurs in exactly one added pair.

The parity lower bound is $8/2 = 4$. In the four selected nonedges each vertex appears exactly once, increasing every degree from three to four, so four insertions suffice and are minimal.

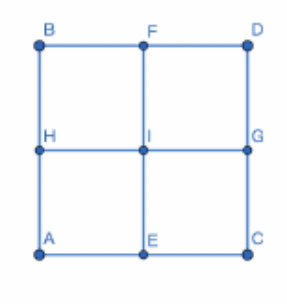
G028

Graphs

2020 Final Part 2 | Question 4(d) | Source page 3 | Difficulty 2
Q-2020-FINAL2-P04-d-decision | Answer provenance: official

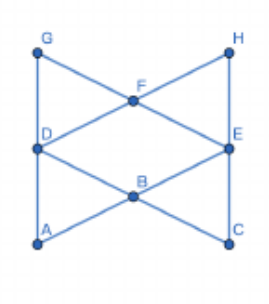
For each of the following graphs, determine whether it has an Euler cycle. If not, insert the least number of edges so that it has an Euler cycle. (You can either draw a picture or write down which edges should be inserted.)

(a) Figure 1



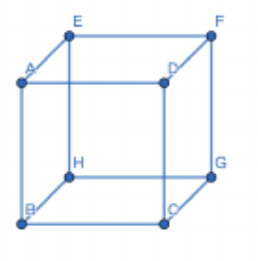
(a) Figure 1

(b) Figure 2



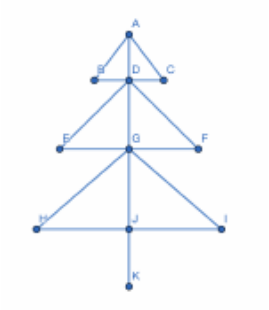
(b) Figure 2

(c) Figure 3



(c) Figure 3

(d) Figure 4



(d) Figure 4

(d) Figure 4 — determine whether it has an Euler cycle.

Official Answer

Source: physical PDF page 3 (2020-final-part-2.pdf).

It is not an Euler circuit, since A, K have odd degree.

Supplementary Explanation

Answer. No.

Derivation/proof. Counting incident edges in the printed figure gives degree 3 at the top vertex A and degree 1 at the bottom vertex K. Every remaining labeled vertex has even degree. Because A and K are odd, the connected graph has an open Euler trail between them but cannot have a closed Euler cycle.

The central vertices D, G each have degree 6; the remaining crossing vertex J has degree four, and the other peripheral vertices have degree two. Only A, K have odd degrees, respectively three and one, excluding an Euler cycle.

G029

Graphs

2020 Final Part 2 | Question 5(a) | Source page 3 | Difficulty 2 | 2 marks
Q-2020-FINAL2-P05-a | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> Suppose we remove a vertex u and all edges incident to u, and get a subgraph G'. Given G' is connected, what is the probability that G is also connected?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

G is also connected if and only if u connects with at least one vertex in G'. Since we know u links to any other vertex with probability p, the probability that it has no edge with other vertices is, $(1-p)^{n-1}$. So the probability that G is connected is $1-(1-p)^{n-1}$.

Supplementary Explanation

Answer. $1-(1-p)^{n-1}$ whenever the conditioning event has positive probability.

Derivation/proof. With G′ connected, G is connected exactly when u has at least one neighbor among the n−1 remaining vertices. The event about G′ depends only on edges not incident to u, so conditioning does not change the independent incident edge probabilities. All n−1 incident edges are absent with probability (1−p)^(n−1). Taking its complement yields the answer.

As usual, the conditioning event must have positive probability; the formula does not define conditioning on a zero-probability event.

G030

Graphs

2020 Final Part 2 | Question 5(b) | Source page 3 | Difficulty 2
Q-2020-FINAL2-P05-b-expectation | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the expectation of X?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

X_e is the indicator random variable that indicates the event, pair of vertices $e = (u, v)$ has an edge. X_e is a Bernoulli trial, thus $E(X_e) = p, V(X_e) = p(1-p)$. There are $\binom{n}{2}$ pairs of vertices in total. So that $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $E[X] = \binom{n}{2}p$.

Derivation/proof. For each of the $\binom{n}{2}$ unordered vertex pairs e let I_e be 1 when its edge exists and 0 otherwise. Then $X = \sum_e I_e$ and $E[I_e] = 1 \cdot p + 0 \cdot (1-p) = p$. By linearity, $E[X] = \sum_e p = \binom{n}{2}p$. Independence is not needed for this expectation step.

There is one indicator for each potential unordered edge. Therefore $E(X) = \sum_e E(I_e) = \binom{n}{2}p$ by linearity, without any need for independence in this particular step.

G031

Graphs

2020 Final Part 2 | Question 5(b) | Source page 3 | Difficulty 2
Q-2020-FINAL2-P05-b-variance | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the variance of X?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

X_e is the indicator random variable that indicates the event, pair of vertices $e = (u, v)$ has an edge. X_e is a Bernoulli trial, thus $E(X_e) = p, V(X_e) = p(1-p)$. There are $\binom{n}{2}$ pairs of vertices in total. So that $E(X) = \binom{n}{2}p$ and $V(X) = \binom{n}{2}p(1-p)$.

Supplementary Explanation

Answer. $\mathrm{Var}(X) = \binom{n}{2}p(1-p)$.

Derivation/proof. Write X as the sum of independent edge indicators I_e. Because $I_e^2 = I_e$, each has variance $E[I_e^2] - E[I_e]^2 = p - p^2 = p(1-p)$. Distinct indicators are independent by the graph model, so their covariances vanish. Adding $\binom{n}{2}$ identical variance terms gives the result.

G032

Graphs

2020 Final Part 2 | Question 5(c) | Source page 3 | Difficulty 2 | 2 marks
Q-2020-FINAL2-P05-c | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> If G has exactly k connected components, what is the minimum possible value of X?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

Suppose that the k connected components are $C_1, C_2, \ldots, C_k$. Since C_i is a connected component with v_i vertices, the minimum number of edges to connect them is $v_i - 1$, which is a tree. The minimum of X is equal to all components reach minimal the same time.

$$\min(X) = \min_{C_i} \sum_{i=1}^{k} \min(C_i) = \min_{C_i} \sum_{i=1}^{k} (v_i - 1) = n - k.$$

Supplementary Explanation

Answer. $n - k$.

Derivation/proof. Let component i have v_i vertices. Connectivity requires at least v_i−1 edges: starting at one vertex, each additional vertex in a spanning traversal needs an edge to the reached set. Summing gives at least $\sum(v_i - 1) = n - k$. Attain this with a tree on each component, including isolated vertices as one-vertex trees. Thus the lower bound is exact.

The component lower bound follows by growing a connected reached set: each newly reached vertex needs a connecting edge. Taking a tree in every component attains all bounds simultaneously.

The component sizes satisfy $\sum_i v_i = n$. A tree in component i uses $v_i - 1$ edges, so the total minimum is $\sum_i (v_i - 1) = n - k$.

G033

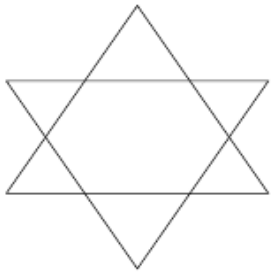
Graphs

2019 Final | Question 13(a) | Source page 16 | Difficulty 2 | 2 marks
Q-2019-FINAL-P13-a | Answer provenance: official

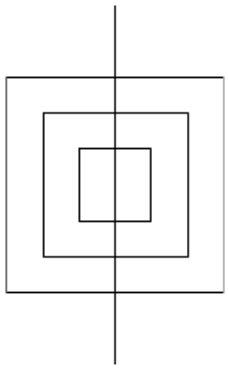
For each of the following graphs, determine whether the graph has an Euler path and/or Euler circuit. Briefly justify your answers.

(a) (b) (c)

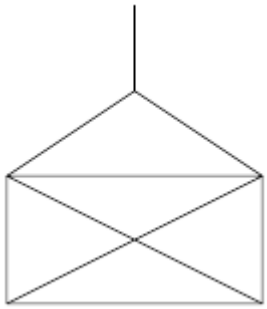
Graph (a):



Graph (b):



Graph (c):



(a)

Official Answer

Source: physical PDF page 16 (2019-final.pdf).

There is an Euler path and Euler circuit because the graph has all vertices of even degree.

Supplementary Explanation

Answer. It has an Euler circuit and therefore also an Euler path.

Derivation/proof. Use the source's connected-intersection interpretation. The six outer tips each meet two edges and have degree two; the six crossings each meet four edge segments and have degree four. The crossings connect the two triangles into one component. Every degree is even, so the connected graph has an Euler circuit. That circuit is also an Euler trail under the exam's convention.

The drawing has no separate vertex dots; the connected-crossing convention used by the printed answer is explicitly retained. Its Euler-path terminology allows a closed traversal.

The six outer tips and six inner crossings exhaust the vertices under the drawing's convention. The graph is connected with every vertex of even degree, so an Euler circuit exists.

G034

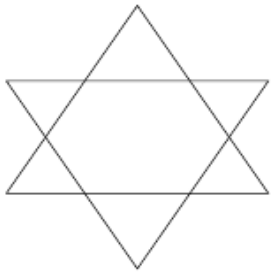
Graphs

2019 Final | Question 13(b) | Source page 16 | Difficulty 2 | 2 marks
Q-2019-FINAL-P13-b | Answer provenance: official

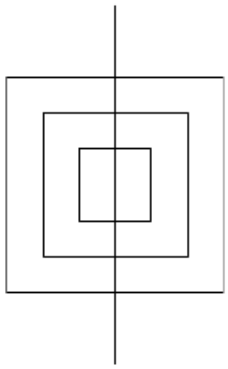
For each of the following graphs, determine whether the graph has an Euler path and/or Euler circuit. Briefly justify your answers.

(a) (b) (c)

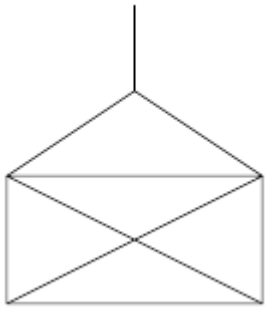
Graph (a):



Graph (b):



Graph (c):



(b)

Official Answer

Source: physical PDF page 16 (2019-final.pdf).

There is an Euler path, but no Euler circuit because the graph has two vertices of odd drgree.

Supplementary Explanation

Answer. It has an Euler path but no Euler circuit.

Derivation/proof. The vertical line connects all three squares. Its upper and lower endpoints each have degree one. Each square corner has degree two, and each of the six line-square crossings has degree four. Thus the graph is connected and has exactly two odd-degree vertices. It admits an open Euler trail between the line endpoints, but an Euler circuit would require every degree even.

The twelve square corners have degree two and the six line-square crossings have degree four. The vertical line makes the graph connected.

Include the twelve square corners and six crossings when counting degrees. There are only two odd-degree vertices, so a trail exists between those endpoints but cannot close to a circuit.

G035

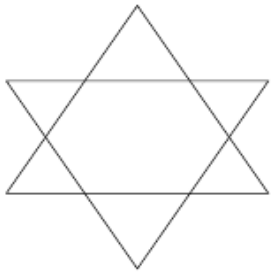
Graphs

2019 Final | Question 13(c) | Source page 16 | Difficulty 2 | 2 marks
Q-2019-FINAL-P13-c | Answer provenance: official

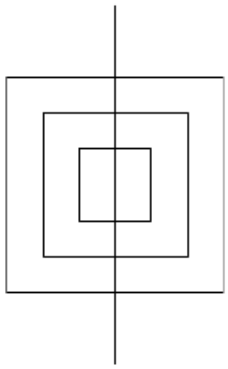
For each of the following graphs, determine whether the graph has an Euler path and/or Euler circuit. Briefly justify your answers.

(a) (b) (c)

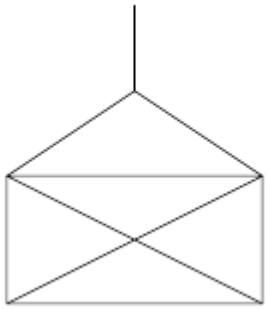
Graph (a):



Graph (b):



Graph (c):



(c)

Official Answer

Source: physical PDF page 16 (2019-final.pdf).

There is no Euler path or Euler circuit because the graph has 4 vertices of odd degree.

Supplementary Explanation

Answer. It has neither an Euler path nor an Euler circuit.

Derivation/proof. The drawing is connected. The chimney's free endpoint has degree one, the roof apex degree three, and each lower corner of the rectangle degree three. The two upper rectangle corners each have degree four, and the central diagonal crossing also has degree four under the source convention. Thus there are four odd-degree vertices. An undirected Euler trail requires zero or two, so neither a trail nor a circuit exists.

The chimney endpoint has degree 1, the roof peak degree 3, and both bottom corners degree 3. The two top corners and central crossing have degree 4.

The chimney's top endpoint has degree 1, the roof apex and both lower corners have degree three. Hence four vertices have odd degree: there are four odd-degree vertices in total, which rules out an Euler trail as well as a circuit.

G036

Graphs

2018 Final | Question 4(c) | Source page 6 | Difficulty 2
Q-2018-FINAL-P04-c-edges | Answer provenance: official

> For this question, no proof is required. Fill in the following recursive definition so that S is the set of complete graphs.
>
> Recursive step: Suppose that $G = (V, E)$ is in S, where $V = \{v_1, v_2, \ldots, v_n\}$. Then $G' = (V', E')$ is in S, where $V' = \{v_1, v_2, \ldots, v_n, v_{n+1}\}$.
>
> (c) $E' =$ ____
>
> [Atomic request: edge set in the recursive step; no proof is required.]

Official Answer

Source: physical PDF page 6 (2018-final.pdf).

Recursive step: $E' = E \cup \{e'_1, e'_2, \ldots, e'_n\}$, where e'_i is the edge between vertex i and $n + 1$.

Supplementary Explanation

Answer. $E' = E \cup \{\{v_i, v_{n+1}\} : 1 \leq i \leq n\}$.

Derivation/proof. Every pair of old vertices is already an edge because G is complete. The only newly introduced pairs contain v_{n+1}, one with each of the n old vertices. Adding exactly those edges and retaining E supplies every distinct vertex pair once and adds no self-loops.

There are exactly n new edges, and the old-old and old-new pairs exhaust all distinct vertex pairs.

G037

Graphs

2018 Final | Question 5(a) | Source page 7 | Difficulty 2
Q-2018-FINAL-P05-a | Answer provenance: official

> (a) For what values of n will a complete graph K_n be bipartite? Explain your answers.

Official Answer

Source: physical PDF page 7 (2018-final.pdf).

Only when $n = 2$. If $n \geq 3$, then consider any vertex colored blue; all of its neighbors will be colored red, and they are neighbors of each other, violating two-coloring. (We do not penalize saying $n = 1$ is bipartite because it is an unclear edge case.)

Supplementary Explanation

Answer. $n = 1$ or $n = 2$, for $n \geq 1$.

Derivation/proof. For K_1, put the lone vertex in one part and let the other part be empty; no edge violates bipartition. For K_2, separate its endpoints. For $n \geq 3$, three vertices form a triangle. If one is blue, both neighbors must be red, but their mutual edge then violates two-coloring. Hence there are no other cases.

The small cases $n = 1, 2$ are both bipartite: a one-vertex graph has the other part empty, and the two-vertex graph places its endpoints in different parts. Here the grading note explicitly accepts the singleton case, which must not be lost when copying the printed main answer.

G038

Graphs

2018 Final | Question 6(a) | Source page 8 | Difficulty 2
Q-2018-FINAL-P06-a | Answer provenance: official

> In this question, we consider simple graphs $G = (V, E)$ with $|V| \geq 3$ where all vertices satisfy $deg(v) \geq 2$. Let us call them two-dense graphs.
>
> Consider the following statement:
>
> All two-dense graphs have a triangle subgraph (a cycle of length 3).
>
> Here is an inductive proof of the statement.
>
> Proof. Let the proposition $P(n)$ be the above statement for $|V| = n$. We will prove $P(n)$ is true for $n \geq 3$.
>
> Base case ($P(3)$): For the two-dense graph G with three vertices, each vertex has a degree of 2, so G is simply a triangle and has itself as a subgraph.
>
> Inductive step: Suppose $P(k)$ is true, i.e. any two-dense graph with k vertices has a triangle subgraph. Extend any such graph G to obtain $G' = (V', E')$ by adding one vertex and at least two new edges from the new vertex to any different vertices in $|V|$. G' is a simple graph with $k + 1$ vertices and all vertices satisfy $deg(v) \geq 2$, so G' is two-dense. G' contains G as a subgraph, so using the inductive assumption, G' also has a triangle subgraph. Therefore, $P(k + 1)$ is true.
>
> (a) Give a counterexample to show that the statement is wrong.

Official Answer

Source: physical PDF page 9 (2018-final.pdf).

Any cycle C_n with $n \geq 4$.

Supplementary Explanation

Answer. C_4 is a counterexample.

Derivation/proof. Take vertices $1, 2, 3, 4$ and edges $12, 23, 34, 41$. This graph is simple, has four vertices, and each vertex has degree two, satisfying the two-dense definition. Any three selected vertices miss at least one edge because there are no diagonals, so there is no triangle. Thus every hypothesis holds while the conclusion fails.

The example lacks edges 13 or 24; every vertex has degree two in its four-cycle. Degree at least two therefore does not force completeness, since nonadjacent pairs remain.

G039

Graphs

2017 Final | Question 13(a) | Source page 17 | Difficulty 2 | 2 marks
Q-2017-FINAL-P13-a | Answer provenance: derived

> In class we learned about several special types of graphs: complete graphs K_n, cycles C_n, and complete bipartite graphs $K_{m,n}$. Recall the definitions:
>
> K_n: For $V = \{v_1, v_2, \ldots, v_n\}$ $(n \geq 1)$, there is exactly one edge between every pair of vertices in V. K_1 is a single vertex and K_2 is two vertices connected by an edge.
>
> C_n: For $V = \{v_1, v_2, \ldots, v_n\}$ $(n \geq 3)$, there is exactly one edge between v_i and v_{i+1} for all $1 \leq i \leq n$, plus exactly one edge from v_n to v_1.
>
> $K_{m,n}$: For every vertex $u_i \in U = \{u_1, u_2, \ldots, u_m\}$ and $v_j \in V = \{v_1, v_2, \ldots, v_n\}$ $(m \geq 1, n \geq 1)$, there is exactly one edge connecting u_i and v_j. There are no edges between two vertices in U, and no edges between two vertices in V.
>
> (a) Can a complete graph K_n be bipartite? Explain what conditions n must satisfy if it is possible.

Official Answer

Source: physical PDF page 17 (2017-final.pdf).

Yes, only if $n = 2$: the graph with two vertices and one edge. Consider the colouring method to determine if a graph is bipartite. If $n > 2$, then if we mark a vertex as red, all other vertices will be marked as blue, and they will be neighbours of each other, so the colouring contradicts the requirement for bipartite graphs.

Supplementary Explanation

Answer. Yes, exactly for $n = 1, 2$ under the standard bipartite definition.

Derivation/proof. For $n = 1$, put the sole vertex in one part and leave the other empty; there are no edges violating the condition. For $n = 2$, put the endpoints in opposite parts. If $n \geq 3$, any three vertices form a triangle. Starting with one color, its two neighbors must both receive the other color, but they are adjacent, which is impossible. A direct enumeration of the small cases agrees.

For the boundary graph K_1, take the unique vertex in one bipartition and the other part empty. It is bipartite despite not belonging to the printed two-vertex-only description.

G040

Graphs

2017 Final | Question 13(b) | Source page 17 | Difficulty 2 | 2 marks
Q-2017-FINAL-P13-b | Answer provenance: official

> In class we learned about several special types of graphs: complete graphs K_n, cycles C_n, and complete bipartite graphs $K_{m,n}$. Recall the definitions:
>
> K_n: For $V = \{v_1, v_2, \ldots, v_n\}$ ($n \geq 1$), there is exactly one edge between every pair of vertices in V. K_1 is a single vertex and K_2 is two vertices connected by an edge.
>
> C_n: For $V = \{v_1, v_2, \ldots, v_n\}$ ($n \geq 3$), there is exactly one edge between v_i and v_{i+1} for all $1 \leq i \leq n$, plus exactly one edge from v_n to v_1.
>
> $K_{m,n}$: For every vertex $u_i \in U = \{u_1, u_2, \ldots, u_m\}$ and $v_j \in V = \{v_1, v_2, \ldots, v_n\}$ ($m \geq 1, n \geq 1$), there is exactly one edge connecting u_i and v_j. There are no edges between two vertices in U, and no edges between two vertices in V.
>
> (b) Can a cycle C_n be bipartite? Explain what conditions n must satisfy if it is possible.

Official Answer

Source: physical PDF page 17 (2017-final.pdf).

Yes, when n is even. Using the colouring method, if vertex 1 is red, then 2 is blue, 3 is red, and so on. Vertex n is blue if n is even, in which case there is no contradiction (all red vertices have blue neighbours, all blue vertices have red neighbours). Vertex n is red if n is odd, but it is a neighbour of vertex 1, so there is a contradiction; no colouring is possible.

Supplementary Explanation

Answer. Exactly when n is even ($n \geq 4$ in this family).

Derivation/proof. Color v_1 red and alternate along $v_1 v_2 \cdots v_n$. If n is even, v_n is blue, so its closing edge to red v_1 is valid and every edge crosses colors. If n is odd, v_n is red, making the closing edge monochromatic. The alternating requirement is forced by every preceding edge, so no other two-coloring can rescue an odd cycle.

Color the cycle alternately: odd-numbered vertices are red and even-numbered vertices are blue. The closing edge has differently colored ends exactly when the length is even.

G041

Graphs

2025 Fall Final | Question 11(a)(ii) | Source page 19 | Difficulty 3 | 1 mark
Q-2025F-FINAL-P11-a-ii | Answer provenance: official

> (a) [3 points, 1 each] Determine whether the following statements are True or False. No argument is needed.
>
> (ii) Consider a simple graph with 8 vertices and exactly 3 edges. This graph can have at least 2 and at most 5 isolated vertices. (An isolated vertex is a vertex of degree 0).

Official Answer

Source: physical PDF page 19 (COMP2711_F2025_Final.pdf).

(ii) True

Supplementary Explanation

Answer. True.

Derivation/proof. Three edges have at most six distinct endpoints, so at least $8 - 6 = 2$ vertices are isolated. Three disjoint edges attain this bound. On the other hand, two nonisolated vertices support at most one simple edge and cannot accommodate three edges. Thus at least three vertices are nonisolated and at most $8 - 3 = 5$ are isolated. A triangle on three vertices attains this second bound. Both bounds are necessary and attainable, so the statement is true.

G042

Graphs

2025 Fall Final | Question 11(a)(iii) | Source page 19 | Difficulty 3 | 1 mark
Q-2025F-FINAL-P11-a-iii | Answer provenance: official

> (a) [3 points, 1 each] Determine whether the following statements are True or False. No argument is needed.
>
> (iii) A simple graph with 10 vertices and 10 edges must contain a cycle.

Official Answer

Source: physical PDF page 19 (COMP2711_F2025_Final.pdf).

(iii) True

Supplementary Explanation

Answer. True.

Derivation/proof. Suppose the ten-vertex graph has no cycle. A finite acyclic component has a vertex of degree at most one: take a longest simple path. An endpoint with a neighbor outside the path would extend it, while another neighbor already on the path would create a cycle. Repeatedly remove a leaf or isolated vertex; induction then shows that each connected component with v_i vertices has exactly $v_i - 1$ edges. With $k \geq 1$ components, the total is $\sum_i (v_i - 1) = 10 - k \leq 9$, contradicting the given ten edges. The graph must therefore contain a cycle.

G043

Graphs

2025 Fall Final | Question 11(b) | Source page 19 | Difficulty 3 | 4 marks
Q-2025F-FINAL-P11-b | Answer provenance: official

> (b) [4 points] Draw the simple graph G with 5 vertices and the following degrees $\{3, 3, 2, 1, 1\}$.

Official Answer

Source: physical PDF page 19 (COMP2711_F2025_Final.pdf), physical PDF page 20 (COMP2711_F2025_Final.pdf).

(b) Here is a simple graph with the following degrees.

page=20 (COMP2711_F2025_Final.pdf)

Supplementary Explanation

Answer. One suitable edge set is $\{AB, AC, BC, AD, BE\}$.

Derivation/proof. Form the triangle on A, B, C with edges AB, AC, BC, giving all three vertices degree two. Add a new vertex D adjacent only to A, and a new vertex E adjacent only to B. Then $\deg(A) = 3, \deg(B) = 3, \deg(C) = 2$, while $\deg(D) = 1$ and $\deg(E) = 1$. This is the requested degree multiset. Every edge joins distinct vertices and no edge is repeated, so the graph is simple. The construction matches the official triangle with two leaves.

```
graph TD
 A --- B
 A --- C
 B --- C
 A --- D
 B --- E
```

G044

Graphs

2025 Spring Final | Question 12(a)(iii) | Source page 16 | Difficulty 3
Q-2025S-FINAL-P12-a-iii | Answer provenance: official

> (a) [3 pts] Consider a simple undirected graph with 6 nodes. Briefly explain (in at most 3 lines) why it is impossible for the nodes to have the following degree sequences:
>
> (iii) 4, 4, 4, 2, 1, 1

Official Answer

Source: physical PDF page 16 (COMP2711_S2025_Final.pdf).

(iii) Note that each of those nodes with degree 4, must be connected to at least one of the nodes with degree 1. (Why?)

So there are at least 3 edges going to those two nodes. So its not possible to have degree 1 for both.

Supplementary Explanation

Answer. Impossible.

Derivation/proof. Let the degree-four vertices be u_1, u_2, u_3, the degree-two vertex be w, and the remaining vertices be v_1, v_2. Each u_i can have at most three neighbors inside $\{u_1, u_2, u_3, w\}$, so it needs at least one edge to $\{v_1, v_2\}$. The three high-degree vertices therefore send at least three edges to those two vertices. But $\deg(v_1) + \deg(v_2) = 2$ provides capacity for at most two, a contradiction. The short answer is: each of three degree-four vertices requires a low-degree neighbor, demanding three incidences from capacity two.

G045

Graphs

2025 Spring Final | Question 12(b) | Source page 16 | Difficulty 3 | 4 marks
Q-2025S-FINAL-P12-b | Answer provenance: official

> (b) [4 pts] How many distinct simple undirected graphs exist on 7 labeled vertices $\{a, b, c, d, e, f, g\}$ such that $\deg(a) \geq 4$?

Official Answer

Source: physical PDF page 16 (COMP2711_S2025_Final.pdf), physical PDF page 17 (COMP2711_S2025_Final.pdf).

(b) First Solution: There are 2^{21} possible graphs in total. Among these, exactly $\binom{6}{3} \times 2^{15}$ graphs satisfy $\deg(a) = 3$; the remaining $2^{21} - \binom{6}{3} \times 2^{15}$ graphs are split equally between those with $\deg(a) \geq 4$ and those with $\deg(a) \leq 2$. This symmetry is due to each graph with $\deg(a) \geq 4$ has a complement graph with $\deg(a) \leq 2$. So, the answer to the question is:

$$\frac{2^{21} - \binom{6}{3} \times 2^{15}}{2} = 720896$$

Second Solution: We can count the number of graphs with $\deg(a) = 4$, $\deg(a) = 5$ and $\deg(a) = 6$ directly:

$$\left(\binom{6}{4} + \binom{6}{5} + \binom{6}{6}\right) \cdot 2^{15} = 720896$$

Supplementary Explanation

Answer. $\left(\binom{6}{4} + \binom{6}{5} + \binom{6}{6}\right)2^{15} = 720896$.

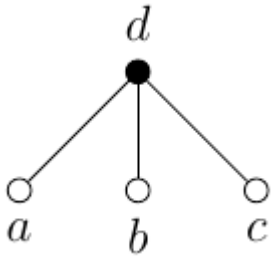
Derivation/proof. If $\deg(a) = d$ for $d = 4, 5, 6$, choose its neighbors from the other six vertices in $\binom{6}{d}$ ways. This fixes all potential edges incident to a. Among the remaining six vertices there are $\binom{6}{2} = 15$ possible edges, each independently present or absent, yielding 2^{15} graphs for each neighbor set. The three degree cases are disjoint, giving $(15 + 6 + 1)2^{15} = 22 \cdot 32768 = 720896$. Vertices are labeled, so distinct neighbor sets or edge sets are distinct graphs without division by isomorphism symmetries.

G046

Graphs

2025 Spring Final | Question 12(c) | Source page 16 | Difficulty 3 | 5 marks
Q-2025S-FINAL-P12-c | Answer provenance: official

(c) [5 pts] Consider a simple undirected graph G with n labeled vertices, where vertex i has degree d_i. Derive an expression for the total number of ordered paths of length 2 in G, in terms of the degrees $d_1, d_2, \ldots, d_n$. For this question, you may use the summation sign Σ to express your answer. Note that the origin and destination of the path must be different. As an example, the ordered paths of length 2 in the graph below are: $\{(adb), (bda), (adc), (cda), (bdc), (cdb)\}$.



Official Answer

Source: physical PDF page 17 (COMP2711_S2025_Final.pdf).

(c) Note that each path consists of 3 nodes. By assuming each node i to be the middle node, the number of possible ordered paths is equal to the number of ways to choose an ordered pair of distinct neighbors of i. Since i has d_i neighbors, the number of ordered pairs of distinct neighbors is $d_i(d_i - 1)$. So the answer to the question is:

$$\sum_{i=1}^{n} d_i(d_i - 1)$$

Also accept:

$$2\sum_{i=1}^{n} \binom{d_i}{2}$$

Supplementary Explanation

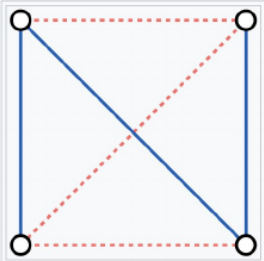
Answer. $\sum_{i=1}^{n} d_i(d_i - 1)$.

Derivation/proof. Fix middle vertex i. There are d_i choices for the starting neighbor and $d_i - 1$ choices for a different ending neighbor, giving $d_i(d_i - 1)$ ordered paths through i. Every length-two path has exactly one middle vertex, so summing over i does not duplicate any ordered path. In the example, the center has degree three and every other vertex degree one. The count is $3 \cdot 2 = 6$, matching the six listed ordered triples.

G047

Graphs

2024 Fall Final | Question 11(c) | Source page 27 | Difficulty 3 | 3 marks
Q-2024F-FINAL-P11-c | Answer provenance: official

> The complement or inverse of a graph G is a graph H on the same vertices such that two distinct vertices of H are adjacent if and only if they are not adjacent in G. A self-complementary graph is a graph which is isomorphic to its complement. As an example, the following graph is self-complementary.
>
> 
>
>
> Assume a graph G is a self-complementary graph with n nodes. Consider H as its complement. Assume S to be the sum of all the nodes degrees in G.
>
> (c) Prove that $n \equiv 0 \pmod 4$ or $n \equiv 1 \pmod 4$. You can use part (b) as a fact even if you could not solve it.

Official Answer

Source: physical PDF page 27 (COMP2711_F2024_Final.pdf).

> (c) Note that $n(n-1) = 2S$ and S is even. So $4 \mid (n)(n-1)$. Also note that between n and $n-1$ one is odd and one is even. So either $4 \mid n$ or $4 \mid n-1$.

Supplementary Explanation

Answer. $n \equiv 0$ or $1 \pmod 4$.

Derivation/proof. By (a), S is even, so $S = 2e$ for an integer e. By (b), $n(n-1) = 2S = 4e$, hence $4 \mid n(n-1)$. Residues zero and one for n give product zero modulo four. If n has residue two, then $n(n-1) \equiv 2 \cdot 1 = 2$. If it has residue three, then $n(n-1) \equiv 3 \cdot 2 = 6 \equiv 2$. The last two cases contradict divisibility by four, leaving only residues zero and one. This proves the necessary condition.

G048

Graphs

2022 Final Part 2 | Question 6(b) | Source page 3 | Difficulty 3 | 4 marks
Q-2022-FINAL2-P06-b | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> If G has exactly m connected components $C_1, C_2, \ldots, C_m$, where C_i is a connected component with v_i vertices, what is the maximum possible value of X?

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (b) For connected component C_i, the maximum number of edges is $\binom{v_i}{2}$, forming a complete graph.
>
> The maximum of X is attained when all components take the maximal value at the same time.
>
> $$\max(X) = \sum_{i=1}^{m} \max(C_i) = \sum_{i=1}^{m} \binom{v_i}{2}.$$

Supplementary Explanation

Answer. $\sum_{i=1}^{m} \binom{v_i}{2}$.

Derivation/proof. No edge joins distinct components. Within the simple component Ci, at most one edge can occur for each unordered pair of its vi vertices, giving at most C(vi,2) edges. Summing these disjoint edge sets bounds X by the displayed value. Attain it by making every component a complete graph K_(vi), which is connected for vi≥1 and creates no cross edges. Hence this is the exact maximum for the supplied component sizes, not a further optimization over those sizes.

The bound is $X \le \sum_{i=1}^{m} \binom{v_i}{2}$. Complete each component to K_{v_i}, including singletons, to attain it. The given sizes cannot be reallocated. The printed answer itself uses a summation despite the paper's general no-summation instruction; that source formatting inconsistency does not change the extremal value.

G049

Graphs

2022 Final Part 2 | Question 6(c) | Source page 4 | Difficulty 3 | 4 marks
Q-2022-FINAL2-P06-c | Answer provenance: official

> There is a random simple undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the probability that G with $n = 4$ vertices has 3 edges and is not connected.

Official Answer

Source: physical PDF page 4 (2022-final-part-2.pdf).

> (c) The simple graph G with 4 vertices and 3 edges is not connected if and only if three edges form a triangle and the last vertex is disconnected, which happens with the probability $\binom{4}{3}p^3(1-p)^3$.

Supplementary Explanation

Answer. $4p^3(1-p)^3$.

Derivation/proof. A disconnected four-vertex simple graph with three edges cannot split2+2, since that allows at most two edges. Its nontrivial component must have three vertices, and all three possible edges must be present: a triangle. Choose the isolated vertex (or triangle vertices) in 4 ways. Each exact configuration has three present and three absent edges among K4's six possibilities, hence probability $p^3(1-p)^3$. These four configurations are disjoint, giving the result.

The structure is a triangle plus an isolated vertex among six potential edges. The four configurations are disjoint. This requires absence of all three edges to the fourth vertex, not merely the presence of a triangle.

G050

Graphs

2021 Final Part 2 | Question Bonus(a) | Source page 4 | Difficulty 3 | 2 marks
Q-2021-FINAL2-BONUS-a | Answer provenance: official

> Let G be an undirected complete graph having n vertices. For a vertex v, $N(v)$ denotes the set of neighbors of v in G. A random walk on G is the following process, which occurs in a sequence of discrete steps: starting at a vertex v_0, we proceed at the first step to a neighbor of v_0 chosen uniformly at random. This may be thought of as choosing a random edge incident on v_0 and walking along it to a vertex $v_1 \in N(v_0)$. At the second step, we proceed to a randomly chosen neighbor of v_1, and so on.
>
> For two given vertices u and v of G, what's the expected number of steps of random walk that begins at u and ends upon first reaching v?

Official Answer

Source: physical PDF page 4 (2021-final-part-2.pdf).

> (a) The probability of visiting v at the next step is $p = \frac{1}{n-1}$ if the current vertex is not v. So the answer is $1/p = n - 1$.

Supplementary Explanation

Answer. $n - 1$ for distinct u, v; 0 if the starting vertex already equals the target.

Derivation/proof. For n≥2 and current vertex different from v, one of its n−1 equally likely neighbors is v. Thus each nonabsorbed step has success probability 1/(n−1), independent of which non-target vertex was reached. Let h be the expected remaining steps. First-step conditioning gives h=1+((n−2)/(n−1))h, hence h=n−1. If u=v, the first hitting time is already zero before any step.

After k steps the non-hitting probability is $(1 - p)^k$, a geometric tail proving finite expectation. Symmetry makes the remaining expectation the same at all non-target vertices. For $n = 2$ the next step hits with certainty, giving expectation one.

The boundary case is zero if $u = v$. Otherwise $h = 1 + \frac{n-2}{n-1}h$, so $h = n - 1$.

G051

Graphs

2020 Final Part 1 | Question 1(a)(i) | Source page 1 | Difficulty 3
Q-2020-FINAL1-P01-a-i | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V(E(u, v) \Leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. Your can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (ii) "Every edge connects a vertex in V_1 and a vertex in V_2".
>
> (iii) "For every vertex in V_1, there are edges that connect it with all vertices in V_2".
>
> "Every vertex in the graph has degree exactly 1".

Official Answer

Source: physical PDF page 1 (2020-final-part-1.pdf).

$\forall u \in V(\exists v \in V(E(u, v) \wedge \forall w \in V(E(u, w) \rightarrow v = w)))$.

Supplementary Explanation

Answer. $\forall u \in V \exists v \in V(E(u, v) \wedge \forall w \in V(E(u, w) \rightarrow w = v))$.

Derivation/proof. For each vertex u, the existential v supplies an adjacent vertex. The universal w clause says that any neighbor of u equals v, excluding a second neighbor. Together they mean degree exactly one. Simplicity excludes loops and multiple edges, so counting distinct neighbors equals degree.

The existence conjunct excludes isolated vertices, while uniqueness excludes a second distinct neighbour.

After requiring an adjacent vertex v, the clause $\forall w \in V(E(u, w) \rightarrow v = w)$ says that every neighbour equals that same v. Together they express existence and uniqueness, excluding isolated vertices as well as vertices with two neighbours.

G052

Graphs

2020 Final Part 2 | Question 5(d) | Source page 4 | Difficulty 3 | 3 marks
Q-2020-FINAL2-P05-d | Answer provenance: official

> There is a random undirected graph G with n vertices, where an edge exists between every pair of vertices with probability p independently. Let X be the random variable denoting the number of edges in the graph.
>
> What is the probability that G with $n = 4$ vertices is connected and X takes its minimal possible value?

Official Answer

Source: physical PDF page 4 (2020-final-part-2.pdf).

G is connected and X is minimal if and only if G is a tree. The probability of having 3 edges in G is $\binom{6}{3}p^3(1-p)^3$. However, if the 3 edges form a triangle, then the last vertex will be disconnected, which happens with probability $\binom{4}{3}p^3(1-p)^3$. So, the final probability is

$$\binom{6}{3}p^3(1-p)^3 - \binom{4}{3}p^3(1-p)^3 = 16p^3(1-p)^3.$$

Supplementary Explanation

Answer. $16p^3(1-p)^3$.

Derivation/proof. A connected four-vertex graph has at least three edges, achieved exactly by trees. There are six potential edges and $\binom{6}{3} = 20$ three-edge subsets. A disconnected three-edge simple graph on four vertices must be a triangle plus an isolate: a 2+2 split permits at most two edges. There are $\binom{4}{3} = 4$ such triangles, leaving 16 trees. Each exact edge set has probability $p^3(1-p)^3$, so multiply by 16.

To verify the disconnected exception is exhaustive, a partition into two pairs permits at most two edges; finer partitions permit no more. Thus a disconnected three-edge graph must consist of a triangle and an isolated vertex.

Of the twenty three-edge subsets, exactly four form a triangle plus an isolated vertex. Removing them leaves $20 - 4 = 16$ trees, each with the same probability $p^3(1-p)^3$.

G053

Graphs

2018 Final | Question 5(b) | Source page 7 | Difficulty 3
Q-2018-FINAL-P05-b | Answer provenance: derived

> (b) For what values of m, n will a complete bipartite graph $K_{m,n}$ have an Euler path but not an Euler circuit? Explain your answers.

Official Answer

Source: physical PDF page 7 (2018-final.pdf).

Only when one of m or n is 2 and the other is odd. In a complete bipartite graph, each of the m vertices have degree n and each of the n vertices have degree m, and we must have exactly two vertices of odd degree.

Supplementary Explanation

Answer. For positive m, n: $m = 2, n$ odd; $n = 2, m$ odd; or $m = n = 1$.

Derivation/proof. Every vertex in the m-part has degree n and every vertex in the n-part has degree m. Since this graph is connected, it needs exactly two odd vertices. Both even gives zero. If n alone is odd, there are m odd vertices, so $m = 2$; the opposite case requires $n = 2$. Both odd gives $m + n$ odd vertices, equaling two precisely at $(1, 1)$. Directly, $K_{1,1}$ is one edge, an open Euler trail with no circuit.

The graph $K_{1,1}$ is a single edge and has an Euler trail. It must be included alongside the larger cases in which the odd-degree count is at most two.

G054

Graphs

2018 Final | Question 6(b) | Source page 8 | Difficulty 3
Q-2018-FINAL-P06-b | Answer provenance: official

> In this question, we consider simple graphs $G = (V, E)$ with $|V| \geq 3$ where all vertices satisfy $deg(v) \geq 2$. Let us call them two-dense graphs.
>
> Consider the following statement:
>
> All two-dense graphs have a triangle subgraph (a cycle of length 3).
>
> Here is an inductive proof of the statement.
>
> Proof. Let the proposition $P(n)$ be the above statement for $|V| = n$. We will prove $P(n)$ is true for $n \geq 3$.
>
> Base case ($P(3)$): For the two-dense graph G with three vertices, each vertex has a degree of 2, so G is simply a triangle and has itself as a subgraph.
>
> Inductive step: Suppose $P(k)$ is true, i.e. any two-dense graph with k vertices has a triangle subgraph. Extend any such graph G to obtain $G' = (V', E')$ by adding one vertex and at least two new edges from the new vertex to any different vertices in $|V|$. G' is a simple graph with $k + 1$ vertices and all vertices satisfy $deg(v) \geq 2$, so G' is two-dense. G' contains G as a subgraph, so using the inductive assumption, G' also has a triangle subgraph. Therefore, $P(k + 1)$ is true.
>
> (b) Since the statement is wrong, the proof must be wrong as well. Choose one of the following statements to explain why the proof is wrong, and explain it.
>
> 1. There is an error in the base case.
> 2. The proof requires another base case, which has not been proven.
> 3. In the inductive step, G' is wrongly constructed; sometimes it is not a two-dense graph.
> 4. In the inductive step, G' may not always have a triangle subgraph.
> 5. In the inductive step, G' is correct and always has a triangle subgraph, but that does not show $P(k + 1)$.

Official Answer

Source: physical PDF page 9 (2018-final.pdf).

The correct choice is (5). Not all graphs with $k + 1$ vertices satisfying $\deg(v) \geq 2$ can be constructed using the procedure in the inductive step. However, $P(k + 1)$ is a statement on all such graphs. Therefore, $P(k + 1)$ being true does not follow from G' containing a triangle.

Supplementary Explanation

Answer. Choice 5.

Derivation/proof. The extension really is two-dense and retains the old triangle. However $P(k + 1)$ quantifies over every two-dense graph with $k + 1$ vertices, not merely the outputs of that extension. For example, C_4 cannot arise by adding a vertex and edges to a triangle, because such an operation never deletes the triangle. The induction step covers only a proper subclass and so cannot establish the universal statement.

The base case and the constructed graph are valid, so choices 1, 3, and 4 misidentify the issue. Adding another base case would not make the construction cover every graph either.

Choose 5, the option identifying the missing universal coverage of the claimed induction extension. The step only constructs a special family; it does not establish the property for all $k + 1$-vertex graphs, so it cannot justify the universal claim.

G055

Graphs

2017 Final | Question 13(c) | Source page 17 | Difficulty 3 | 3 marks
Q-2017-FINAL-P13-c | Answer provenance: derived

> In class we learned about several special types of graphs: complete graphs K_n, cycles C_n, and complete bipartite graphs $K_{m,n}$. Recall the definitions:
>
> K_n: For $V = \{v_1, v_2, \ldots, v_n\}$ $(n \geq 1)$, there is exactly one edge between every pair of vertices in V. K_1 is a single vertex and K_2 is two vertices connected by an edge.
>
> C_n: For $V = \{v_1, v_2, \ldots, v_n\}$ $(n \geq 3)$, there is exactly one edge between v_i and v_{i+1} for all $1 \leq i \leq n$, plus exactly one edge from v_n to v_1.
>
> $K_{m,n}$: For every vertex $u_i \in U = \{u_1, u_2, \ldots, u_m\}$ and $v_j \in V = \{v_1, v_2, \ldots, v_n\}$ $(m \geq 1, n \geq 1)$, there is exactly one edge connecting u_i and v_j. There are no edges between two vertices in U, and no edges between two vertices in V.
>
> (c) Can a complete bipartite graph $K_{m,n}$ have an Euler path but not an Euler circuit? Explain what conditions m and n must satisfy if it is possible.

Official Answer

Source: physical PDF page 17 (2017-final.pdf).

For a bipartite graph, the vertices in U all have degree n and the vertices in V all have degree m. We know that there exists an Euler path but not an Euler circuit if there are exactly two vertices of odd degree and all other vertices have even degree. For such a scenario to occur, this means that either $m = 2$ and n is odd, or $n = 2$ and m is odd. So it is possible.

Supplementary Explanation

Answer. Yes: $m = 2$ and n odd, or $n = 2$ and m odd, or $(m, n) = (1, 1)$.

Derivation/proof. The graph is connected for $m, n \geq 1$. Its m vertices in one part have degree n, and its other n vertices have degree m. An open Euler trail requires exactly two odd-degree vertices. If both parameters are even, the count is zero. If only n is odd, the count is m, requiring $m = 2$; symmetrically the other mixed case needs $n = 2$. If both are odd, the count is $m + n$, which equals two exactly for $m = n = 1$. Checking the single edge independently confirms the final case.

The single-edge graph $K_{1,1}$ has exactly two odd-degree vertices and therefore an Euler trail, supplying the small case omitted by the printed classification.

G056

Graphs

2017 Final | Question 14(c) | Source page 18 | Difficulty 3 | 4 marks
Q-2017-FINAL-P14-c | Answer provenance: official

> In this question, we want to find the minimum number of edges in a connected graph with n vertices.
>
> (c) Using the above, prove that there is no connected graph with n vertices and fewer than $n - 1$ edges.

Official Answer

Source: physical PDF page 18 (2017-final.pdf).

Suppose in contradiction that G is a connected graph that has n vertices and k edges where $k < n - 1$. Then using part (b), we have a connected graph that has $n - 1$ vertices and $k - 1$ edges. We can repeat this process until we reach a connected graph with at least 2 vertices and 0 edges. This is a contradiction, as such a graph is clearly not connected.

Supplementary Explanation

Answer. Every connected n-vertex graph has at least $n - 1$ edges.

Derivation/proof. The $n = 1$ case holds since edge counts are nonnegative. Suppose $n \geq 2$ and a connected graph has $k < n - 1$ edges. Connectivity gives every degree at least one. The degree sum $2k < 2n$ ensures some vertex has degree one. Delete it and its incident edge; a simple path between remaining vertices cannot use a degree-one vertex internally, so connectivity is preserved. Repeat while edges remain. After k deletions, at least $n - k > 1$ vertices remain and no edges remain, impossible for a connected graph. The reduction is used only for at least two vertices, its valid domain.

At each deletion, the difference between the vertex count and edge count stays $n - k > 1$, so the reduction is invoked only within its valid range.

If a graph with n vertices and $k < n$ edges had every vertex of degree at least two, the handshake identity would instead give $\sum \deg v \geq 2n$. But $\sum \deg v = 2k < 2n$, a contradiction; some vertex has degree at most one.

G057

Graphs

2016 Midterm | Question 5(a) | Source page 9 | Difficulty 3 | 5 marks
Q-2016-MID-P05-a | Answer provenance: official

> There are n dots painted on a paper. A kid draws some line segments on the paper and each line segment connects exactly two dots as the two end-points of this line segment. For example, the following figure shows 3 line segments and 6 dots.
>
> Note that if one line segment (connecting two dots) is an extension of another line segment (connecting two dots where one of them is the dot connected by the previous line segment mentioned), we will still refer them as two separate line segments (not one line segment). For example, the following figure shows 3 line segments and 6 dots.
>
> We say that the n dots are all connected if for every pair of two dots x and y, there exists a sequence of line segments, namely $(d_1, d_2), (d_2, d_3), (d_3, d_4), \ldots, (d_{l-1}, d_l)$ where d_i is a dot for $i \in [1, l]$, such that $d_1 = x$ and $d_l = y$. For example, the following figure shows that the 6 dots are all connected but the above 2 figures shows that 6 dots are not all connected.
>
> The following shows an example that the line segments form a loop.
>
> (a) Prove by contraposition that if the kid only draws $n - 2$ line segments, then the n dots are not all connected.

Official Answer

Source: PDF physical page 9 (2016-midterm.pdf).

(a) Let p be the kid only draws $n - 2$ line segments.

Let q be the n dots are not all connected.

To prove $p \Rightarrow q$ using contraposition, we state $\neg q$ which is "n dots are all conntected". Since all n dots are connected, each dot must have a least one line segment attached to it. We use a highlighter pen to highlight one arbitrary line segment and the two dots connected by the line segment. Then we highlight another line segment such that it connects one of the "highlighted" dots and one of the "unhighlighted" dots. Thus we will have two line segments and three dots highlighted and $n - 3$ dots unhighligthed. The process repeats until all n dots are highlighted. The process is repeatable since every dot is connected. That means there must be some unhighlighted dots connecting with a highlighted dot at any state of the process. After n dots are highlighted, there are $n - 1$ line segments highlighted. That implies the kid has drawn at least $n - 1$ line segments ($\neg p$).

Supplementary Explanation

Answer. With only $n - 2$ segments, the dots cannot all be connected.

Derivation/proof. Assume the graph is connected. Start with one highlighted vertex. While fewer than n vertices are highlighted, a path to an unhighlighted vertex has a first edge leaving the highlighted set. Select that edge and its new endpoint. Each step adds one previously unhighlighted vertex using a distinct edge, so reaching all n vertices selects $n - 1$ edges. Thus a connected graph cannot have only $n - 2$ edges. This proves the contrapositive for the meaningful range $n \geq 2$.

For $n = 1$, the antecedent of drawing -1 segments is impossible, so the statement is vacuous. For $n \geq 2$, each chosen crossing edge is new because it reaches a previously unhighlighted vertex.

The construction selects $n - 1$ distinct edges, one for each newly reached vertex after the starting vertex.

G058

Graphs

2016 Midterm | Question 5(b) | Source page 9 | Difficulty 3 | 5 marks
Q-2016-MID-P05-b | Answer provenance: official

> There are n dots painted on a paper. A kid draws some line segments on the paper and each line segment connects exactly two dots as the two end-points of this line segment. For example, the following figure shows 3 line segments and 6 dots.
>
> Note that if one line segment (connecting two dots) is an extension of another line segment (connecting two dots where one of them is the dot connected by the previous line segment mentioned), we will still refer them as two separate line segments (not one line segment). For example, the following figure shows 3 line segments and 6 dots.
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> We say that the n dots are all connected if for every pair of two dots x and y, there exists a sequence of line segments, namely $(d_1, d_2), (d_2, d_3), (d_3, d_4), \ldots, (d_{l-1}, d_l)$ where d_i is a dot for $i \in [1, l]$, such that $d_1 = x$ and $d_l = y$. For example, the following figure shows that the 6 dots are all connected but the above 2 figures shows that 6 dots are not all connected.
>
> The following shows an example that the line segments form a loop.
>
> (b) Prove by contradiction that if the kid draws a loop that connects the n dots, then s/he needs at least n line segments to draw the loop.

Official Answer

Source: PDF physical page 9 (2016-midterm.pdf).

(b) Assume that it is possible to draw a loop with smaller than n line segments for n dots. Since each dot needs to be connected with exactly two line segments in a loop, we count the total number of connections between a line segment and a dot, there are $2n$ connections. If the loop has smaller than n line segment, say, there are only k line segments where $k < n$, and thus each line segment on average needs to connect $2n/k$ dots (more than 2 dots). However, each line segment can exactly connect two dots. By contradiction, it requires at least n line segments to draw a loop that connects n dots.

Supplementary Explanation

Answer. At least n line segments are needed.

Derivation/proof. Suppose such a loop has $m < n$ segments. Every one of the n cycle vertices has two incident loop edges, so there are $2n$ endpoint incidences. Each segment has exactly two endpoints and contributes two incidences, giving only $2m < 2n$. This contradiction establishes the bound (a simple cycle has exactly n edges).

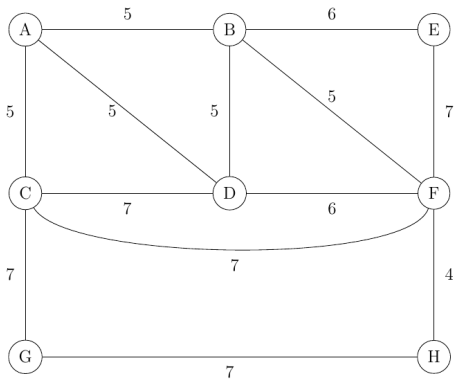
Equivalently, $2n = 2m < 2n$ is impossible. Count the loop's own edges; extra chords do not reduce how many edges its loop needs.

G059

Graphs

2025 Fall Final | Question 11(c) | Source page 19 | Difficulty 4 | 5 marks
Q-2025F-FINAL-P11-c | Answer provenance: official

(c) [5 points] In the following graph, find the length of the shortest tour so that the postman can traverse every edge at least once and return to the starting point.



Official Answer

Source: physical PDF page 20 (COMP2711_F2025_Final.pdf).

(c) The graph is not Eulerian since two vertices have odd degrees (A, F). To make it Eulerian, we need to add the minimum weight of duplicate edges to pair up the odd-degree vertices. The shortest path between AF is $A \to B \to F$ with costs of 5+5.

The shortest tour is:

$$H \to G \to C \to F \to D \to C \to A \to D \to B \to A \to B \to F \to E \to B \to F \to H$$

Total weight calculation:

$$\text{Total} = 7 \cdot 5 + 4 + 2 \cdot 6 + 5 \cdot 7 = 86.$$

Thus, the shortest tour length is 86.

Supplementary Explanation

Answer. The shortest closed tour has length 86.

Derivation/proof. Read the graph edge by edge: $AB = AC = AD = BD = BF = 5$, $BE = DF = 6$, $CD = CF = CG = EF = GH = 7$ and $FH = 4$. The total original edge weight is $5 \cdot 5 + 2 \cdot 6 + 5 \cdot 7 + 4 = 76$. Vertex degrees are $A : 3, B : 4, C : 4, D : 4, E : 2, F : 5, G : 2, H : 2$; only A, F are odd. A closed tour enters and leaves each vertex equally often. Consequently its repeated-edge multigraph must have exactly A, F odd. The component containing A must also contain F, because each component has an even number of odd-degree vertices. Thus the repeated edges contain an A–F path.

There is no direct AF edge. Two-edge paths through B, C, D have lengths $5 + 5 = 10$, $5 + 7 = 12$ and $5 + 6 = 11$. Any path of at least three edges has length at least 12, since every edge weighs at least four. The shortest path is therefore ABF, of weight ten. Every closed tour has length at least $76 + 10 = 86$.

Duplicate AB and BF. Every degree is now even and the graph remains connected, so an Euler circuit exists. The explicit tour $H, G, C, F, D, C, A, D, B, A, B, F, E, B, F, H$ uses every original edge at least once and repeats exactly AB, BF. Its weight is 86, attaining the lower bound and proving optimality.

G060

Graphs

2021 Final Part 2 | Question Bonus(b) | Source page 4 | Difficulty 4 | 8 marks
Q-2021-FINAL2-BONUS-b | Answer provenance: official

> Let G be an undirected complete graph having n vertices. For a vertex v, $N(v)$ denotes the set of neighbors of v in G. A random walk on G is the following process, which occurs in a sequence of discrete steps: starting at a vertex v_0, we proceed at the first step to a neighbor of v_0 chosen uniformly at random. This may be thought of as choosing a random edge incident on v_0 and walking along it to a vertex $v_1 \in N(v_0)$. At the second step, we proceed to a randomly chosen neighbor of v_1, and so on.
>
> Redo (a) for the case where G is a line graph of n vertices, and u is at one end and v is at the other end.

Official Answer

Source: physical PDF page 5 (2021-final-part-2.pdf).

> (b) Number of the vertices as $1, \ldots, n$. Let T_i denote the number of steps to go from i to n. We have $T_n = 0$, so $E[T_n] = 0$. The goal is to find $E[T_1]$.
>
> From vertex 1, we can only go right, so $T_1 = 1 + T_2$ and $E[T_1] = 1 + E[T_2]$. Rewrite it as $E[T_1] - E[T_2] = 1$. For any $1 < i < n$, we go left or right with equal probability, so $E[T_i] = 1 + \frac{1}{2}E[T_{i-1}] + \frac{1}{2}E[T_{i+1}]$. Rearrange it as $E[T_i] - E[T_{i+1}] = 2 + E[T_{i-1}] - E[T_i]$. Now we have
>
> $$\begin{aligned} E[T_1] - E[T_2] &= 1 \\ E[T_2] - E[T_3] &= 3 \\ &\cdots \\ E[T_{n-1}] - E[T_n] &= 2n - 3. \end{aligned}$$
>
> Summing up, we have $E[T_1] = E[T_n] + 1 + 3 + \cdots + 2n - 3 = (n-1)^2$.

Supplementary Explanation

Answer. $(n-1)^2$.

Derivation/proof. Label the path vertices 1,...,n and let h_i be expected steps to reach n. Then h_n=0 and h_1=1+h_2. For 1<i<n, h_i=1+(h_(i−1)+h_(i+1))/2. Define d_i=h_i−h_(i+1); rearrangement gives d_1=1 and d_i=d_(i−1)+2, hence d_i=2i−1. Telescoping yields h_1−h_n=1+3+...+(2n−3)=(n−1)². Expectations are finite: from any vertex there is a positive uniformly bounded chance to follow right steps to n within n−1 steps, giving a geometric block tail. For n=1 the answer is zero.

For $n \geq 2$, from any non-target vertex, following rightward steps reaches the target within $n-1$ steps with conditional probability at least $2^{-(n-1)} > 0$. Each block has this uniform success lower bound, so a geometric tail controls the hitting time. Explicitly,

$$h_1 = h_1 - h_n = \sum_{i=1}^{n-1} d_i = 1 + 3 + \cdots + (2n-3) = (n-1)^2.$$

For $n = 1$, both endpoints coincide and the expectation is zero.

The difference recurrence is $d_i = d_{i-1} + 2$, giving $d_i = 2i - 1$.

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